Project Title

NEWS & SOCIAL MEDIA ANALYTICS ENGINE

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DECLARATION

The Mini Project Report entitled "**News & Social Media Analytics Engine"** is a record of Bonafide work of **K.MRUDULA – 2520090073., B.NAVYA – 2520090189.**submitted in partial fulfillment for the award of B. Tech in Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other University.

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CERTIFICATE

This is to certify that the mini project based report entitled "**Project Title"** is a bonafide work done and submitted by **K.MRUDULA – 2520090073., B.NAVYA – 2520090189.** in partial

fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department Computer Science and Information Technology K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD

Signature of the External Examiner

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1.ABSTRACT

The **News & Social Media Analytics Engine** is a Java-based application designed to organize, process, and analyze news-related textual information. With the rapid growth of digital news and social media content, manually searching for important information, identifying popular topics, and understanding public opinions can be time-consuming. This project provides a simple interface to manage news posts and analyze their content.
The system allows users to add news posts with a title, category, source, content, and optional news URL. It provides category-wise and keyword-based searching to help users find relevant news efficiently. The application uses an ArrayList to store news posts and a HashMap to maintain keyword frequency. Searching techniques are used to retrieve matching posts, while sorting is used to arrange trending keywords according to their frequency.
The system also performs basic sentiment classification by checking predefined positive and negative keywords and categorizes the content as positive, negative, or neutral. A web-based dashboard displays news posts, sentiment counts, and trending topics. Users can open the complete content of a news post and access its original URL when provided.
The project demonstrates the practical application of fundamental data structures and algorithms in text-based news analytics. It can be further enhanced with live news APIs, advanced Natural Language Processing, database integration, and graphical visualizations.

2.INTRODUCTION

In today's digital world, news websites and social media platforms generate a large amount of textual information every day. Finding relevant news and identifying popular topics manually can require considerable time and effort. The News & Social Media Analytics Engine provides a simple application for organizing, searching, and analyzing news-related content.

The project is implemented using Java for the backend and HTML, CSS, and JavaScript for the user interface. The application allows users to add news posts, search by title or category, view complete news content, and open the original news link when available.

The project applies important Data Structures and Algorithms concepts. ArrayList is used to maintain the collection of news posts, HashMap is used to count keyword frequencies, linear searching retrieves matching news items, and sorting identifies the most frequent keywords. A basic keyword-based sentiment analysis method classifies posts into positive, negative, and neutral categories.

The main purpose of this project is to demonstrate how data structures and algorithms can be applied to a practical text-analysis application. The current implementation serves as a prototype that can be extended with real-time data collection, advanced sentiment analysis, persistent storage, and more detailed analytics.

3.LITERATURE SURVEY

3.1 Google News

Google News organizes news articles from multiple publishers and presents stories based on topics and user interests. It demonstrates how categorization and information retrieval help users find relevant news. Our project implements a simpler version of category-based news searching.

3.2 NewsAPI

NewsAPI provides access to news articles through an API. It demonstrates how applications can retrieve structured news information from external sources. This project can integrate a news API in the future to replace manually added and sample posts with externally collected news.

3.3 Natural Language Toolkit (NLTK)

NLTK provides tools for processing and analyzing human-language text. It supports tasks such as tokenization and text analysis. Our prototype uses basic keyword processing and can be extended with an NLP library for more advanced analysis.

3.4 VADER Sentiment Analysis

VADER is a lexicon- and rule-based sentiment analysis tool designed for text. It demonstrates how sentiment can be estimated from words and linguistic patterns. Our current application uses a simpler predefined list of positive and negative keywords to classify news content.

3.5 Apache OpenNLP

Apache OpenNLP provides tools for natural language processing, including tokenization and other text-processing tasks. Such tools could improve the text-analysis capabilities of the proposed system in future versions.

3.6 Keyword Frequency Analysis

Keyword frequency analysis identifies words that occur frequently in a collection of documents. Frequent keywords can help users discover topics that appear repeatedly. Our application uses a HashMap to count keywords and sorting to display the most frequent terms as trending topics.

These existing platforms and techniques provide useful ideas for news organization, text processing, sentiment analysis, and keyword identification. The proposed project

combines simplified versions of these ideas with data structures and algorithms in a single application.

4.Problem Statement and Objectives

Problem Statement:

Managing, searching, and analyzing large collections of news-related textual information manually is time-consuming and makes it difficult to identify relevant posts and frequently occurring topics.

Problem Description:

Users need a simple system to organize news posts, search by category or keyword, view complete article content, identify frequent keywords, and understand basic sentiment classifications**.**

Objectives:

1. To develop a Java-based news analytics application with a simple web interface.
2. To store and manage news posts using an ArrayList.
3. To use a HashMap to calculate keyword frequencies.
4. To implement searching based on titles, categories, sources, and content.
5. To sort keywords according to their frequency and identify trending topics.
6. To classify news content as positive, negative, or neutral using predefined keywords.
7. To allow users to add posts and view their complete content.
8. To provide an option to open the original news URL when available.
9. To demonstrate the practical use of data structures and algorithms in a real-world application.

5.HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- **Computer/Laptop:** A computer capable of running Java and a web browser.
- **Processor:** Intel or equivalent processor.
- **RAM:** 4 GB or more recommended.
- **Storage:** Sufficient free storage for the source code and project files.
- **Internet:** Required for accessing external news URLs; the local prototype can run without continuously collecting online data.

Software Requirements:

- **Operating System:** Windows, Linux, or another Java-supported operating system.
- **Programming Language:** Java.
- **Java Development Kit:** JDK 17 or later.
- **Code Editor:** Visual Studio Code.
- **Frontend Technologies:** HTML, CSS, and JavaScript.
- **Backend Technology:** Java HTTP Server.
- **Web Browser:** Google Chrome or another modern browser.
- **Data Structures:** ArrayList and HashMap.
- **Algorithms:** Linear searching, sorting, keyword frequency counting, and basic sentiment classification.

Technologies Used : Java handles the application logic and HTTP requests. HTML provides the structure of the dashboard, CSS controls the visual presentation, and JavaScript manages user interactions and communicates with the Java backend through HTTP endpoints.

6.IMPLEMENTATION

- **6.1 Module 1:** News Post Management
  This module stores news posts using an ArrayList. Each post contains an ID, title, category, source, content, URL, and sentiment. Users can add new posts through the web interface.
- **6.2 Module 2:** Searching
  The searching module uses linear search to check whether the entered keyword appears in the title, category, source, or content. Matching posts are returned and displayed to the user.
- **6.3 Module 3:** Trending Topic Identification
  The system uses a HashMap to maintain the frequency of keywords extracted from news titles, categories, and content. The entries are copied into a list and sorted in descending order of frequency. The most frequent keywords are displayed as trending topics.
- **6.4 Module 4:** Sentiment Analysis
  The sentiment module checks news content against predefined positive and negative keywords. It assigns a Positive, Negative, or Neutral label based on the keyword counts. This is a basic rule-based approach and not a trained machine-learning model.
- **6.5 Web Interface**
  The dashboard displays the total number of posts, sentiment counts, search field, news-post cards, trending keywords, and a form for adding news. Users can select a post to read its full content and open the original article URL when one is available.
- **6.6 Execution Procedure**
  1. Save the source code as NewsAnalytics.java.
  2. Open the project folder in Visual Studio Code.
  3. Open the terminal in the same folder.
  4. Compile the program using javac NewsAnalytics.java.
  5. Start the application using java NewsAnalytics.
  6. Open http://localhost:8080 in a browser.
  7. Test the search, add-post, sentiment, trending-topic, and full-content features.

7.RESULTS & DISCUSSION

The application provides a working dashboard for demonstrating news management and basic text analytics.

7.1 Dashboard Output

The dashboard displays the total number of news posts and counts for positive, negative, and neutral sentiment labels.

Screenshot to insert: Capture the home page showing the dashboard and news posts.

7.2 Category-Wise Search

Users can search for terms such as Sports, Technology, Health, or Education. The system displays matching posts based on their titles, categories, sources, or content.

Screenshot to insert: Search for "Sports" and capture the matching results.

7.3 Add News Post

The application allows users to add a new post by entering its title, category, source, content, and optional news URL. The new post appears in the news list.

Screenshot to insert: Capture the Add News Post form and a successfully added post.

7.4 Full News Content

Selecting a news title or the Read Full News button opens a view containing the complete post content. When a URL is available, the user can open the associated external webpage.

Screenshot to insert: Capture the full-news view.

7.5 Trending Topics

The system calculates keyword frequencies and sorts them in descending order. Frequently occurring keywords appear in the Trending Topics section.

Screenshot to insert: Capture the trending keywords displayed on the dashboard.

7.6 Discussion

The project demonstrates the practical use of ArrayList, HashMap, linear searching, and sorting. The dashboard makes the sample data easy to explore.

Sentiment results are based on predefined keywords and should be interpreted as a basic demonstration rather than a highly accurate sentiment prediction.

8.CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The News & Social Media Analytics Engine demonstrates how data structures and algorithms can be used to organize and analyze news content. The project implements news post management using ArrayList, keyword frequency counting using HashMap, linear searching for finding relevant posts, and sorting for ranking trending keywords.

The application also provides a basic keyword-based sentiment classification feature and a web dashboard for displaying news information. Users can add posts, search by category or keyword, view complete content, and open associated news URLs.

The project meets its main objective of demonstrating a simple news analytics application using Java and fundamental data structures and algorithms. Its current implementation is a prototype that can be extended with more advanced features.

8.2 Future Scope

1. **Live News Integration:** Connect news APIs to retrieve updated articles automatically.
2. **Database Integration:** Store news posts permanently using MongoDB or another database.
3. **Advanced Sentiment Analysis:** Use trained machine-learning or NLP models for more accurate classification.
4. **Improved Visualizations:** Add charts and graphs for sentiment distribution and keyword trends.
5. **Multilingual Support:** Extend analysis to different languages.
6. **Historical Trend Analysis:** Store data over time and compare topic frequencies across different periods.
7. **User Features:** Add authentication, user preferences, and filtering options.

8. **Performance Improvements:** Optimize searching and sorting for larger collections of news posts.

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